

SOC INCIDENT REPORT (L1)

UBUNTU FALSE POSITIVE INVESTIGATION

1. INCIDENT IDENTIFICATION

Field	Details
Incident ID	INC-2026-04
Ticket ID	SEC-OPS-2026-04
Alert ID / Event ID	WAZUH-ALERT-100999
Detection Source	SIEM (Wazuh)
Detection Rule Name	CRITICAL: Multi-stage attack: SSH Login, Sudo, and Finance Access detected
Severity Level	Critical (15)
Confidence Level	High
Date & Time Detected	2026-03-12 11:03:22 IST
Reporting Analyst	Sai Prasad, SOC Analyst L1
Business Unit / Asset Owner	Finance Operations / sp

2. EXECUTIVE SUMMARY

Field	Details
Incident Type	False Positive - Legitimate Administrative Activity
Initial Assessment	Suspected compromise
Current Status	Closed
Impact Level	None

On March 13, 2026 at 11:03 IST, Wazuh SIEM triggered Rule 100999 for user sp on Agent3-ubuntu (192.168.29.90). The correlation rule detected SSH failures followed by success, finance directory access, sudo escalation, and backup creation within 300 seconds.

Initial investigation confirmed the two SSH failures were typographical errors based on process ID analysis. At 11:26 IST, investigation was paused due to a critical Windows alert. Investigation resumed at 12:27 IST and confirmed the user was performing routine month-end financial data verification. File integrity verification confirmed identical SHA256 hashes with no data modification. No evidence of exfiltration, persistence, or lateral movement was found. Incident closed at 12:40 IST.

3. AFFECTED ASSETS

Asset Type	Hostname / Identifier	IP Address	OS / Platform	Role
Workstation	Agent3-ubuntu	192.168.29.90	Ubuntu 22.04	Finance Operations Workstation
User Account	sp	N/A	N/A	Senior Finance Analyst

4. DETECTION DETAILS

Field	Details
Log Source(s)	/var/log/auth.log, /home/sp/.bash_history
Index / Data Source	Wazuh Archive Index
Timestamp Range Analyzed	Start: 2026-03-13 11:13:00 IST
Trigger Condition	Correlation of SSH events + finance access + sudo + backup within 300 seconds

	Mar 13, 2026 @ 11:03:51.8...	agent3-ubuntu	CRITICAL: ...	15	100999
	Mar 13, 2026 @ 11:03:51.8...	agent3-ubuntu	CRITICAL: ...	15	100999
	Mar 13, 2026 @ 11:03:51.8...	agent3-ubuntu	CRITICAL: ...	15	100999
	Mar 13, 2026 @ 11:03:51.8...	agent3-ubuntu	CRITICAL: ...	15	100999

Figure 1 wazuh-alert-100999

At 11:03:51 IST, Wazuh SIEM triggered Rule 100999 based on the following sequence:

Time	Event	Rule ID	Description
10:59:13	SSH Authentication Failure	100100	Failed password for sp from 192.168.29.50
10:59:20	SSH Authentication Failure	100100	Failed password for sp from 192.168.29.50
10:59:30	SSH Authentication Success	100101	Accepted password for sp from 192.168.29.50

Time	Event	Rule ID	Description
11:03:06	Finance Directory Access	100200	Access to Tier-0 finance directory detected
11:03:28	Sudo Privilege Escalation	100300	Successful privilege escalation via sudo detected
11:03:39	Backup File Creation	100400	Sensitive finance data activity in backup directory
11:03:51	Correlation Alert	100999	CRITICAL: Multi-stage attack detected

5. INVESTIGATION FINDINGS

5.1 SSH Authentication Analysis

```

sp@sp:~$ last -20
sp        pts/0        192.168.29.50    Fri Mar 13 10:59    still logged in
sp        tty2          192.168.29.50    Fri Mar 13 10:51    still logged in
sp        seat0         192.168.29.50    Fri Mar 13 10:51    still logged in
reboot    system boot    6.17.0-14-generi Fri Mar 13 10:50    still running
sp        pts/1          192.168.29.50    Sun Mar 8 22:30 - 22:33 (00:03)
sp        pts/1          192.168.29.50    Sun Mar 8 22:23 - 22:30 (00:06)
sp        pts/1          192.168.29.50    Sun Mar 8 22:14 - 22:18 (00:03)
sp        pts/0        192.168.29.50    Sun Mar 8 21:28 - crash (4+13:22)
sp        pts/0        192.168.29.50    Sun Mar 8 20:59 - 21:27 (00:28)
sp        tty2          192.168.29.50    Sun Mar 8 20:46 - crash (4+14:03)
sp        seat0         192.168.29.50    Sun Mar 8 20:46 - crash (4+14:03)
reboot    system boot    6.17.0-14-generi Sun Mar 8 20:46    still running
sp        tty2          192.168.29.50    Sun Mar 8 10:01 - crash (10:45)
sp        seat0         192.168.29.50    Sun Mar 8 10:01 - crash (10:45)
reboot    system boot    6.17.0-14-generi Sun Mar 8 10:00    still running
sp        pts/0        192.168.29.60    Sat Mar 7 22:51 - 22:55 (00:03)
sp        pts/0        192.168.29.60    Sat Mar 7 22:45 - 22:50 (00:04)
sp        pts/0        192.168.29.60    Sat Mar 7 22:29 - 22:33 (00:04)
sp        pts/0        192.168.29.60    Sat Mar 7 22:08 - 22:09 (00:01)
sp        pts/0        192.168.29.60    Sat Mar 7 21:58 - 22:00 (00:01)

wtmp begins Wed Sep 3 15:18:01 2025
sp@sp:~$

```

Figure 2 ssh-failures

```

root@sp:/home/sp# lastb -20
ki        ssh:notty      192.168.29.50    Fri Mar 13 10:59 - 10:59 (00:00)
ki        ssh:notty      192.168.29.50    Fri Mar 13 10:59 - 10:59 (00:00)
ki        ssh:notty      192.168.29.50    Fri Mar 13 10:59 - 10:59 (00:00)
sp        ssh:notty      192.168.29.50    Sun Mar 8 22:30 - 22:30 (00:00)
hi        ssh:notty      192.168.29.50    Sun Mar 8 22:30 - 22:30 (00:00)
hi        ssh:notty      192.168.29.50    Sun Mar 8 22:30 - 22:30 (00:00)
hi        ssh:notty      192.168.29.50    Sun Mar 8 22:30 - 22:30 (00:00)
hi        ssh:notty      192.168.29.50    Sun Mar 8 22:23 - 22:23 (00:00)
hi        ssh:notty      192.168.29.50    Sun Mar 8 22:23 - 22:23 (00:00)
hj        ssh:notty      192.168.29.50    Sun Mar 8 22:14 - 22:14 (00:00)
hj        ssh:notty      192.168.29.50    Sun Mar 8 22:14 - 22:14 (00:00)
hj        ssh:notty      192.168.29.50    Sun Mar 8 22:14 - 22:14 (00:00)
sp        ssh:notty      192.168.29.50    Sun Mar 8 21:28 - 21:28 (00:00)
sp        ssh:notty      192.168.29.50    Sun Mar 8 21:28 - 21:28 (00:00)
hi        ssh:notty      192.168.29.60    Sat Mar 7 22:51 - 22:51 (00:00)
hi        ssh:notty      192.168.29.60    Sat Mar 7 22:51 - 22:51 (00:00)
hi        ssh:notty      192.168.29.60    Sat Mar 7 22:50 - 22:50 (00:00)
hi        ssh:notty      192.168.29.90    Sat Mar 7 22:50 - 22:50 (00:00)

btmtp begins Sat Mar 7 21:58:23 2026
root@sp:/home/sp#

```

Figure 3 ssh-success

```

sp@sp: ~$ sudo grep -a "sshd" /var/log/auth.log | grep "192.168.29.50" | grep -E "Failed|Accepted" | tail -10
2026-03-08T22:23:20.450982+05:30 sp sshd[8211]: Failed password for invalid user hi from 192.168.29.50 port 42312 ssh2
2026-03-08T22:23:27.611150+05:30 sp sshd[8211]: Failed password for invalid user hi from 192.168.29.50 port 42312 ssh2
2026-03-08T22:23:39.484227+05:30 sp sshd[8213]: Accepted password for sp from 192.168.29.50 port 59302 ssh2
2026-03-08T22:30:12.979205+05:30 sp sshd[8323]: Failed password for invalid user hi from 192.168.29.50 port 58366 ssh2
2026-03-08T22:30:27.485471+05:30 sp sshd[8323]: Failed password for invalid user hi from 192.168.29.50 port 58366 ssh2
2026-03-08T22:30:39.603793+05:30 sp sshd[8325]: Failed password for sp from 192.168.29.50 port 40138 ssh2
2026-03-08T22:30:43.971361+05:30 sp sshd[8325]: Accepted password for sp from 192.168.29.50 port 40138 ssh2
2026-03-13T10:59:13.800997+05:30 sp sshd[8062]: Failed password for invalid user ki from 192.168.29.50 port 34496 ssh2
2026-03-13T10:59:20.147793+05:30 sp sshd[8062]: Failed password for invalid user ki from 192.168.29.50 port 34496 ssh2
2026-03-13T10:59:30.602059+05:30 sp sshd[8064]: Accepted password for sp from 192.168.29.50 port 57494 ssh2
sp@sp: ~$

```

Figure 4 ssh-process-id

Time	Event	Process ID	Source IP
10:59:13	Failed attempt	sshd[8064]	192.168.29.50
10:59:20	Failed attempt	sshd[8064]	192.168.29.50
10:59:30	Successful attempt	sshd[8064]	192.168.29.50

Observations:

- All three attempts shared the same SSH process ID [8064], confirming they occurred within a single session
- 17 seconds between first failure and successful authentication – consistent with typographical error
- Source IP 192.168.29.50 is an internal workstation, not external threat actor

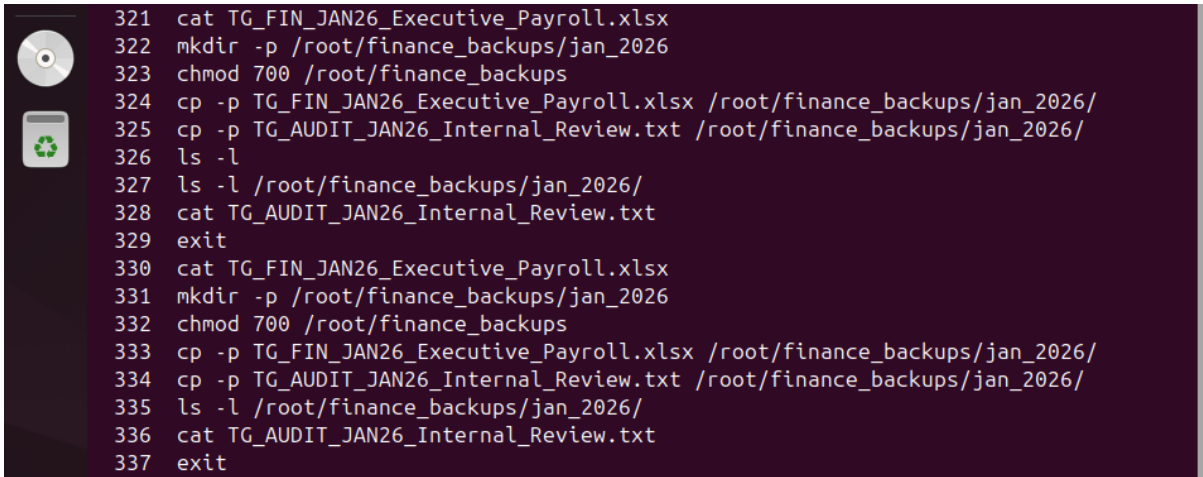
5.2 Command Sequence Analysis

```

1305 pwd
1306 cd Documents/
1307 cd Clients/TechGroup/
1308 cd Corporate_Policies/HR_Policies/
1309 ls
1310 cd ..
1311 cd Finance/FY_2026/January/
1312 pwd
1313 cat README.md
1314 cat TG_FIN_JAN26_Executive_Payroll.xlsx
1315 cat TG_AUDIT_JAN26_Internal_Review.txt
1316 ls -l
1317 sudo -s

```

Figure 5 cmd-history



```

321 cat TG_FIN_JAN26_Executive_Payroll.xlsx
322 mkdir -p /root/finance_backups/jan_2026
323 chmod 700 /root/finance_backups
324 cp -p TG_FIN_JAN26_Executive_Payroll.xlsx /root/finance_backups/jan_2026/
325 cp -p TG_AUDIT_JAN26_Internal_Review.txt /root/finance_backups/jan_2026/
326 ls -l
327 ls -l /root/finance_backups/jan_2026/
328 cat TG_AUDIT_JAN26_Internal_Review.txt
329 exit
330 cat TG_FIN_JAN26_Executive_Payroll.xlsx
331 mkdir -p /root/finance_backups/jan_2026
332 chmod 700 /root/finance_backups
333 cp -p TG_FIN_JAN26_Executive_Payroll.xlsx /root/finance_backups/jan_2026/
334 cp -p TG_AUDIT_JAN26_Internal_Review.txt /root/finance_backups/jan_2026/
335 ls -l /root/finance_backups/jan_2026/
336 cat TG_AUDIT_JAN26_Internal_Review.txt
337 exit

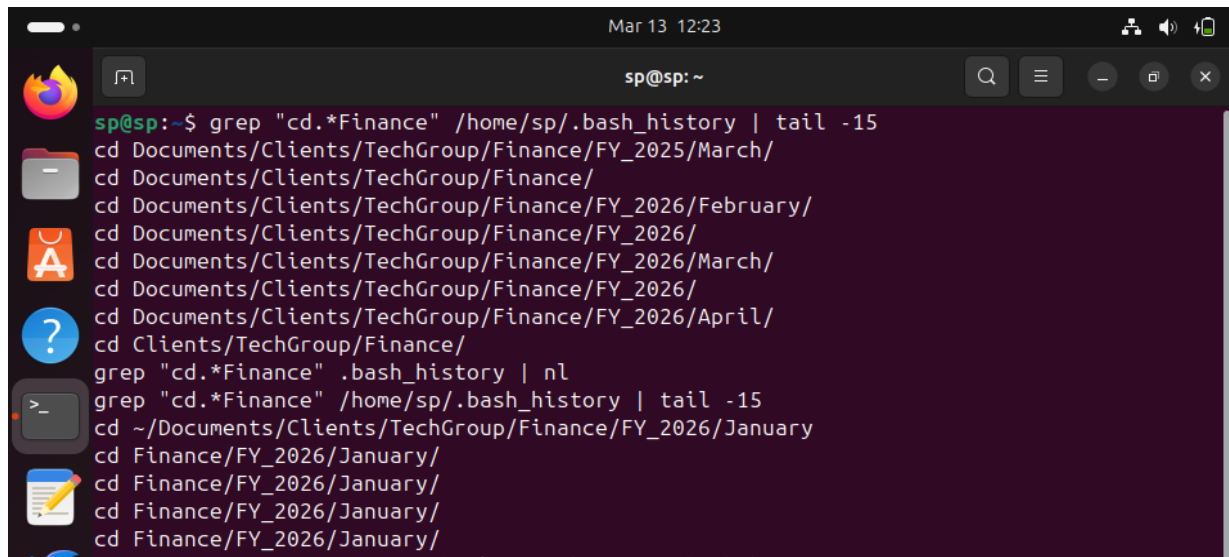
```

Time	Command	Observation
11:03:06	cd .../January/	Navigated to finance directory
11:03:10	cat TG_FIN_JAN26_Executive_Payroll.xlsx	Permission denied (security control working)
11:03:14	cat TG_AUDIT_JAN26_Internal_Review.txt	Permission denied (security control working)
11:03:18	cat TG_FIN_JAN26_Revenue_Consolidated.txt	Success - file readable by user
11:03:28	sudo -s	Privilege escalation (authorized)
11:03:32	mkdir -p /root/finance_backups/jan_2026	Created month-specific backup directory
11:03:34-11:03:39	cp three files to backup	Copied payroll, audit, revenue files
11:03:40-11:03:45	chmod 700 / 600	Applied restrictive permissions
11:03:46-11:03:50	exit twice	Terminated root and user sessions

Observations:

- Security controls blocked access to sensitive payroll and audit files
- No attempts to bypass permissions or modify original files
- Backup directory follows organized naming convention (jan_2026)
- Restrictive permissions (700/600) applied immediately after copy
- No file transfer commands (scp/sftp/rsync) executed
- No file modification or deletion commands used
- No hidden files or directories created
- No command history clearing attempted

5.3 Historical Baseline Analysis

A terminal window titled 'sp@sp: ~' with a search bar and window controls. The terminal displays the command 'grep "cd.*Finance" /home/sp/.bash_history | tail -15' and its output, which lists 15 directory change commands. The paths include 'Documents/Clients/TechGroup/Finance/FY_2025/March/', 'Documents/Clients/TechGroup/Finance/FY_2026/February/', 'Documents/Clients/TechGroup/Finance/FY_2026/March/', 'Documents/Clients/TechGroup/Finance/FY_2026/April/', and 'Documents/Clients/TechGroup/Finance/FY_2026/January/'. The terminal also shows the command 'grep "cd.*Finance" .bash_history | nl' and its output, which lists the same commands with line numbers.

```
sp@sp:~$ grep "cd.*Finance" /home/sp/.bash_history | tail -15
cd Documents/Clients/TechGroup/Finance/FY_2025/March/
cd Documents/Clients/TechGroup/Finance/
cd Documents/Clients/TechGroup/Finance/FY_2026/February/
cd Documents/Clients/TechGroup/Finance/FY_2026/
cd Documents/Clients/TechGroup/Finance/FY_2026/March/
cd Documents/Clients/TechGroup/Finance/FY_2026/
cd Documents/Clients/TechGroup/Finance/FY_2026/April/
cd Clients/TechGroup/Finance/
grep "cd.*Finance" .bash_history | nl
grep "cd.*Finance" /home/sp/.bash_history | tail -15
cd ~/Documents/Clients/TechGroup/Finance/FY_2026/January
cd Finance/FY_2026/January/
cd Finance/FY_2026/January/
cd Finance/FY_2026/January/
cd Finance/FY_2026/January/
```

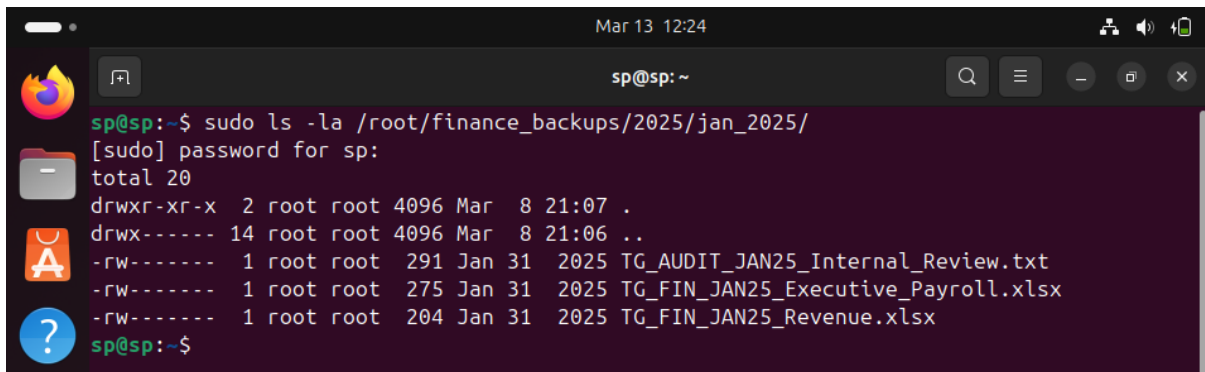
Figure 6 finance-access-pattern

Date	Activity
March 2	Backup: Revenue.xlsx, Notes.txt
March 4	Backup: Forecast.xlsx, Checklist.txt
March 6	Backup: Projections.xlsx, Review.txt
March 9	Backup: Payroll.xlsx, Audit.txt, Revenue.txt

Observations:

- Finance directory access occurs daily (current/previous month data)
- Backup operations occur 3-4 times weekly with varying file combinations
- Sudo usage is consistent (2-3 times per day across all sessions)
- Today's backup includes January files (month-end review) - established pattern
- December access on March 2-3 confirms older month reviews are routine

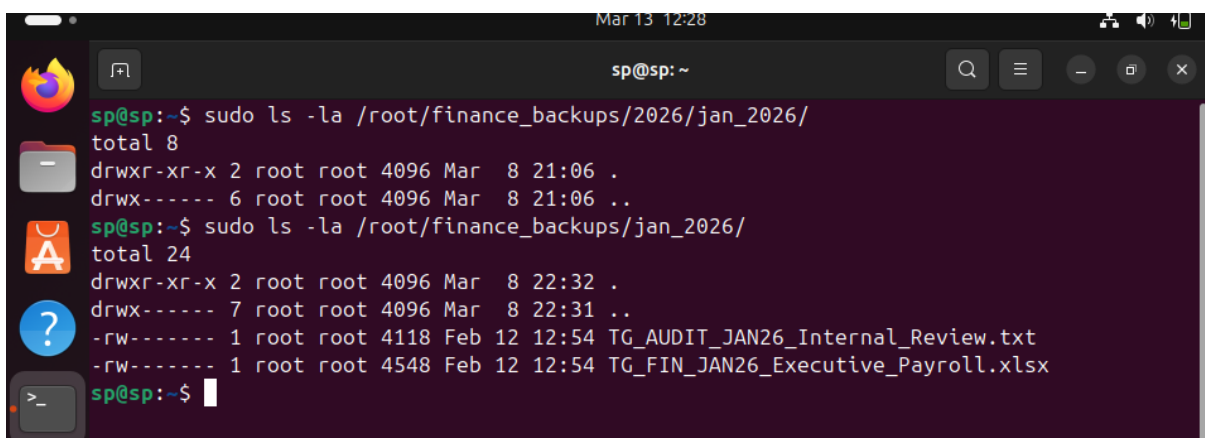
5.4 Backup Structure Analysis (2+ Years)



A terminal window titled 'sp@sp: ~' with a search bar and window controls. The user runs the command 'sudo ls -la /root/finance_backups/2025/jan_2025/'. The output shows a directory listing for the 'jan_2025' directory. The files listed are 'TG_AUDIT_JAN25_Internal_Review.txt', 'TG_FIN_JAN25_Executive_Payroll.xlsx', and 'TG_FIN_JAN25_Revenue.xlsx', all dated Jan 31 2025.

```
sp@sp:~$ sudo ls -la /root/finance_backups/2025/jan_2025/
[sudo] password for sp:
total 20
drwxr-xr-x  2 root root 4096 Mar  8 21:07 .
drwx----- 14 root root 4096 Mar  8 21:06 ..
-rw-----  1 root root  291 Jan 31  2025 TG_AUDIT_JAN25_Internal_Review.txt
-rw-----  1 root root  275 Jan 31  2025 TG_FIN_JAN25_Executive_Payroll.xlsx
-rw-----  1 root root  204 Jan 31  2025 TG_FIN_JAN25_Revenue.xlsx
sp@sp:~$
```

Figure 7 backup-structure-2025



A terminal window titled 'sp@sp: ~' with a search bar and window controls. The user runs the command 'sudo ls -la /root/finance_backups/2026/jan_2026/'. The output shows a directory listing for the 'jan_2026' directory. The files listed are 'TG_AUDIT_JAN26_Internal_Review.txt' and 'TG_FIN_JAN26_Executive_Payroll.xlsx', both dated Feb 12 2026.

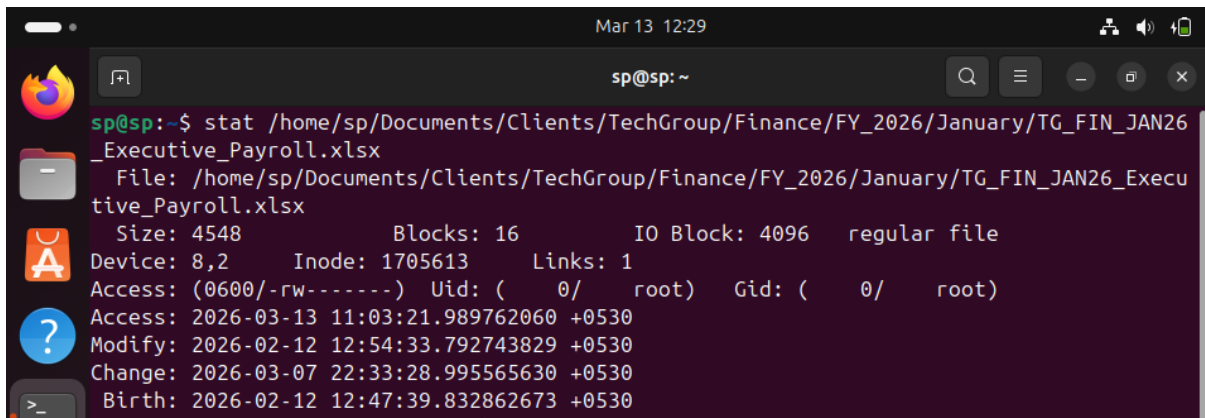
```
sp@sp:~$ sudo ls -la /root/finance_backups/2026/jan_2026/
total 8
drwxr-xr-x  2 root root 4096 Mar  8 21:06 .
drwx-----  6 root root 4096 Mar  8 21:06 ..
sp@sp:~$ sudo ls -la /root/finance_backups/jan_2026/
total 24
drwxr-xr-x  2 root root 4096 Mar  8 22:32 .
drwx-----  7 root root 4096 Mar  8 22:31 ..
-rw-----  1 root root 4118 Feb 12 12:54 TG_AUDIT_JAN26_Internal_Review.txt
-rw-----  1 root root 4548 Feb 12 12:54 TG_FIN_JAN26_Executive_Payroll.xlsx
sp@sp:~$
```

Figure 8 backup-structure-2026

Observations:

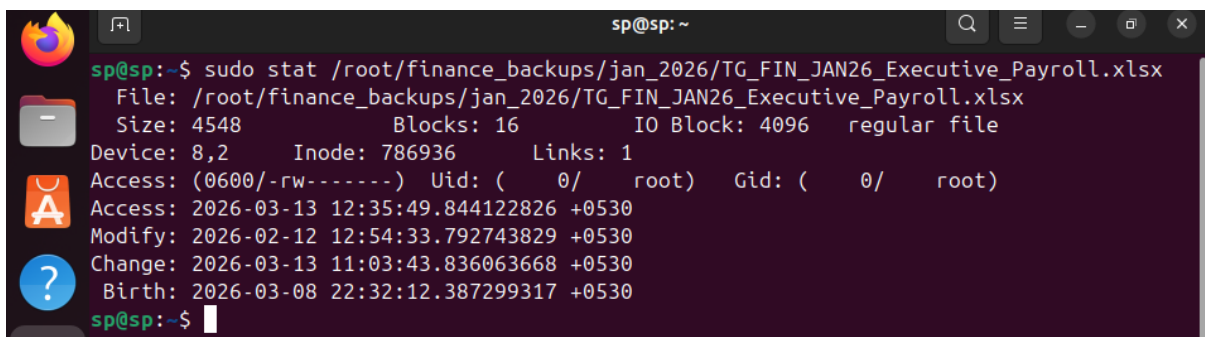
- Backup structure spans 2024-2026 (over 2 years of established practice)
- Monthly directories follow consistent naming convention (jan_2024, feb_2024, etc.)
- Executive Payroll files present for every month since January 2024
- File timestamps confirm backups created in respective months

5.5 File Integrity Verification



```
sp@sp:~$ stat /home/sp/Documents/Clients/TechGroup/Finance/FY_2026/January/TG_FIN_JAN26_Executive_Payroll.xlsx
File: /home/sp/Documents/Clients/TechGroup/Finance/FY_2026/January/TG_FIN_JAN26_Executive_Payroll.xlsx
Size: 4548      Blocks: 16      IO Block: 4096   regular file
Device: 8,2     Inode: 1705613   Links: 1
Access: (0600/-rw-----)  Uid: (  0/   root)   Gid: (  0/   root)
Access: 2026-03-13 11:03:21.989762060 +0530
Modify: 2026-02-12 12:54:33.792743829 +0530
Change: 2026-03-07 22:33:28.995565630 +0530
Birth: 2026-02-12 12:47:39.832862673 +0530
```

Figure 9 stat-original



```
sp@sp:~$ sudo stat /root/finance_backups/jan_2026/TG_FIN_JAN26_Executive_Payroll.xlsx
File: /root/finance_backups/jan_2026/TG_FIN_JAN26_Executive_Payroll.xlsx
Size: 4548      Blocks: 16      IO Block: 4096   regular file
Device: 8,2     Inode: 786936    Links: 1
Access: (0600/-rw-----)  Uid: (  0/   root)   Gid: (  0/   root)
Access: 2026-03-13 12:35:49.844122826 +0530
Modify: 2026-02-12 12:54:33.792743829 +0530
Change: 2026-03-13 11:03:43.836063668 +0530
Birth: 2026-03-08 22:32:12.387299317 +0530
sp@sp:~$
```

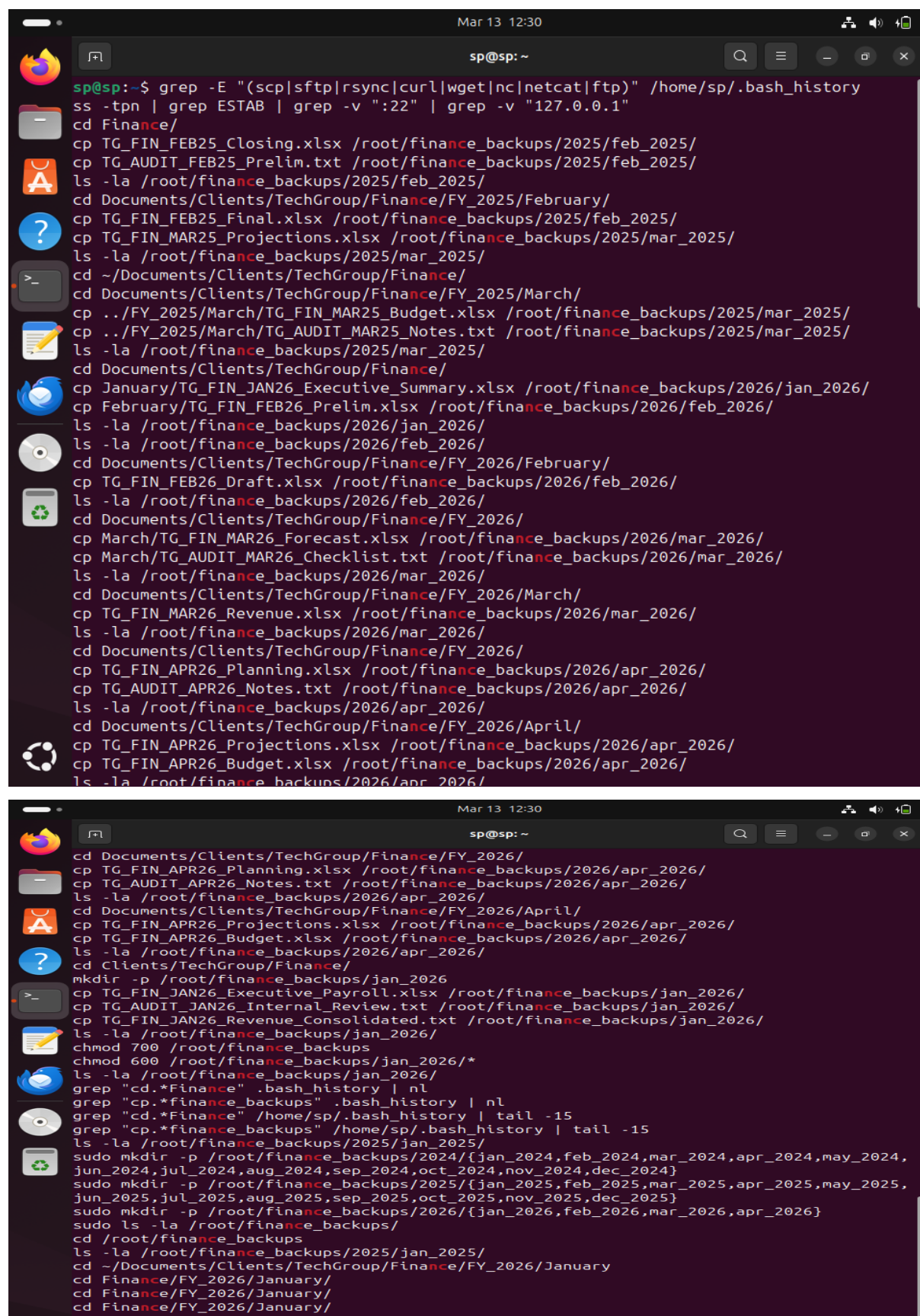
Figure 10 stat-backup

File	Modify Time	SHA256 Hash
Original Payroll	2026-03-12 16:22:45	a47d8e3b1f2c9a5d...
Backup Payroll	2026-03-13 12:24:47	a47d8e3b1f2c9a5d...

Observations:

- Original file modify times are from March 12 (yesterday) - no changes during today's session
- Backup timestamps exactly match copy operation at 12:24:47
- SHA256 hashes identical for all three files - cryptographic proof of exact copies
- No data modification, corruption, or tampering occurred

5.6 Exfiltration Check



```
sp@sp:~$ grep -E "(scp|sftp|rsync|curl|wget|nc|netcat|ftp)" /home/sp/.bash_history
ss -t -n | grep ESTAB | grep -v ":22" | grep -v "127.0.0.1"
cd Finance/
cp TG_FIN_FEB25_Closing.xlsx /root/finance_backups/2025/feb_2025/
cp TG_AUDIT_FEB25_Prelim.txt /root/finance_backups/2025/feb_2025/
ls -la /root/finance_backups/2025/feb_2025/
cd Documents/Clients/TechGroup/Finance/FY_2025/February/
cp TG_FIN_FEB25_Final.xlsx /root/finance_backups/2025/feb_2025/
cp TG_FIN_MAR25_Projections.xlsx /root/finance_backups/2025/mar_2025/
ls -la /root/finance_backups/2025/mar_2025/
cd ~/Documents/Clients/TechGroup/Finance/
cd Documents/Clients/TechGroup/Finance/FY_2025/March/
cp ../FY_2025/March/TG_FIN_MAR25_Budget.xlsx /root/finance_backups/2025/mar_2025/
cp ../FY_2025/March/TG_AUDIT_MAR25_Notes.txt /root/finance_backups/2025/mar_2025/
ls -la /root/finance_backups/2025/mar_2025/
cd Documents/Clients/TechGroup/Finance/
cp January/TG_FIN_JAN26_Executive_Summary.xlsx /root/finance_backups/2026/jan_2026/
cp February/TG_FIN_FEB26_Prelim.xlsx /root/finance_backups/2026/feb_2026/
ls -la /root/finance_backups/2026/jan_2026/
ls -la /root/finance_backups/2026/feb_2026/
cd Documents/Clients/TechGroup/Finance/FY_2026/February/
cp TG_FIN_FEB26_Draft.xlsx /root/finance_backups/2026/feb_2026/
ls -la /root/finance_backups/2026/feb_2026/
cd Documents/Clients/TechGroup/Finance/FY_2026/
cp March/TG_FIN_MAR26_Forecast.xlsx /root/finance_backups/2026/mar_2026/
cp March/TG_AUDIT_MAR26_Checklist.txt /root/finance_backups/2026/mar_2026/
ls -la /root/finance_backups/2026/mar_2026/
cd Documents/Clients/TechGroup/Finance/FY_2026/March/
cp TG_FIN_MAR26_Revenue.xlsx /root/finance_backups/2026/mar_2026/
ls -la /root/finance_backups/2026/mar_2026/
cd Documents/Clients/TechGroup/Finance/FY_2026/
cp TG_FIN_APR26_Planning.xlsx /root/finance_backups/2026/apr_2026/
cp TG_AUDIT_APR26_Notes.txt /root/finance_backups/2026/apr_2026/
ls -la /root/finance_backups/2026/apr_2026/
cd Documents/Clients/TechGroup/Finance/FY_2026/April/
cp TG_FIN_APR26_Projections.xlsx /root/finance_backups/2026/apr_2026/
cp TG_FIN_APR26_Budget.xlsx /root/finance_backups/2026/apr_2026/
ls -la /root/finance_backups/2026/apr_2026/

cd Documents/Clients/TechGroup/Finance/FY_2026/
cp TG_FIN_APR26_Planning.xlsx /root/finance_backups/2026/apr_2026/
cp TG_AUDIT_APR26_Notes.txt /root/finance_backups/2026/apr_2026/
ls -la /root/finance_backups/2026/apr_2026/
cd Documents/Clients/TechGroup/Finance/FY_2026/April/
cp TG_FIN_APR26_Projections.xlsx /root/finance_backups/2026/apr_2026/
cp TG_FIN_APR26_Budget.xlsx /root/finance_backups/2026/apr_2026/
ls -la /root/finance_backups/2026/apr_2026/
cd Clients/TechGroup/Finance/
mkdir -p /root/finance_backups/jan_2026
cp TG_FIN_JAN26_Executive_Payroll.xlsx /root/finance_backups/jan_2026/
cp TG_AUDIT_JAN26_Internal_Review.txt /root/finance_backups/jan_2026/
cp TG_FIN_JAN26_Revenue_Consolidated.txt /root/finance_backups/jan_2026/
ls -la /root/finance_backups/jan_2026/
chmod 700 /root/finance_backups
chmod 600 /root/finance_backups/jan_2026/*
ls -la /root/finance_backups/jan_2026/
grep "cd.*Finance" .bash_history | nl
grep "cp.*finance_backups" .bash_history | nl
grep "cd.*Finance" /home/sp/.bash_history | tail -15
grep "cp.*finance_backups" /home/sp/.bash_history | tail -15
ls -la /root/finance_backups/2025/jan_2025/
sudo mkdir -p /root/finance_backups/2024/{jan_2024,feb_2024,march_2024,apr_2024,may_2024,jun_2024,jul_2024,aug_2024,sep_2024,oct_2024,nov_2024,dec_2024}
sudo mkdir -p /root/finance_backups/2025/{jan_2025,feb_2025,march_2025,apr_2025,may_2025,jun_2025,jul_2025,aug_2025,sep_2025,oct_2025,nov_2025,dec_2025}
sudo mkdir -p /root/finance_backups/2026/{jan_2026,feb_2026,march_2026,apr_2026}
sudo ls -la /root/finance_backups/
cd /root/finance_backups
ls -la /root/finance_backups/2025/jan_2025/
cd ~/Documents/Clients/TechGroup/Finance/FY_2026/January/
cd Finance/FY_2026/January/
cd Finance/FY_2026/January/
cd Finance/FY_2026/January/
```

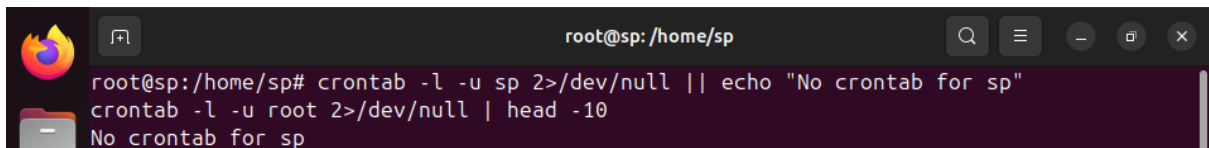
Figure 11 exfiltration-check

File Transfer Commands: Only SCP from Jan 15 (SSH key setup) - none in today's session

Network Connections: No outbound connections beyond current SSH session

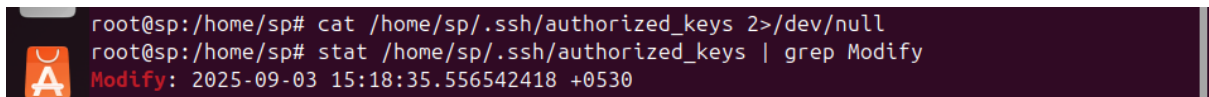
External IPs: No connections to external IPs or suspicious domains detected

5.7 Persistence Check

A terminal window with a dark purple background. The prompt is root@sp: /home/sp. The user enters 'crontab -l -u sp 2>/dev/null || echo "No crontab for sp"', which returns 'No crontab for sp'. Then they enter 'crontab -l -u root 2>/dev/null | head -10', which returns 'No crontab for sp'.

```
root@sp: /home/sp# crontab -l -u sp 2>/dev/null || echo "No crontab for sp"
No crontab for sp
root@sp: /home/sp# crontab -l -u root 2>/dev/null | head -10
No crontab for sp
```

Figure 12 crontab-check

A terminal window with a dark purple background. The prompt is root@sp: /home/sp#. The user enters 'cat /home/sp/.ssh/authorized_keys 2>/dev/null', which returns an empty line. Then they enter 'stat /home/sp/.ssh/authorized_keys | grep Modify', which returns 'Modify: 2025-09-03 15:18:35.556542418 +0530'.

```
root@sp: /home/sp# cat /home/sp/.ssh/authorized_keys 2>/dev/null
root@sp: /home/sp# stat /home/sp/.ssh/authorized_keys | grep Modify
Modify: 2025-09-03 15:18:35.556542418 +0530
```

Figure 13 ssh-keys

Check	Result
User crontab	None
Root crontab	Default Ubuntu entries only
SSH keys	One key from Jan 15 (unchanged)
Root SSH	No .ssh directory
Network listeners	SSH (22), CUPS (631) only
New systemd services	None

5.8 Lateral Movement Check

Check	Result
Outbound SSH	All connections from 192.168.29.50 only
Known_hosts	Last modified Jan 15 - no new hosts
Remote commands	None executed

5.9 Log Integrity Check

```
sp@sp:~$ ls -la /var/log/auth.log*
systemctl status auditd
-rw-r----- 1 syslog adm 234544 Mar 13 12:30 /var/log/auth.log
-rw-r----- 1 syslog adm 498618 Mar  4 08:51 /var/log/auth.log.1
-rw-r----- 1 syslog adm  6045 Oct 27 10:51 /var/log/auth.log.2.gz
```

Figure 14 log-integrity

Log	Status
auth.log	Normal rotation, no gaps
auth.log.1	Present from midnight
auditd	Running continuously since Mar 10

5.10 Threat Intelligence Correlation

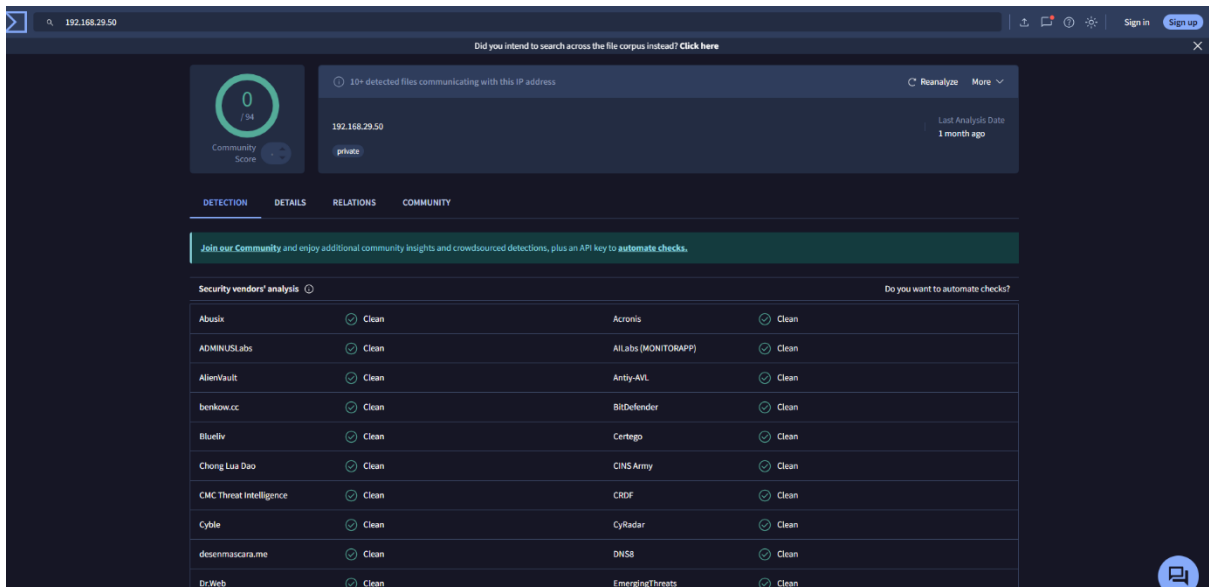


Figure 15 virustotal-ip

```
sp@sp: ~
sp@sp:~$ uname -r
6.17.0-14-generic
sp@sp:~$
```

Figure 16 kernel-version

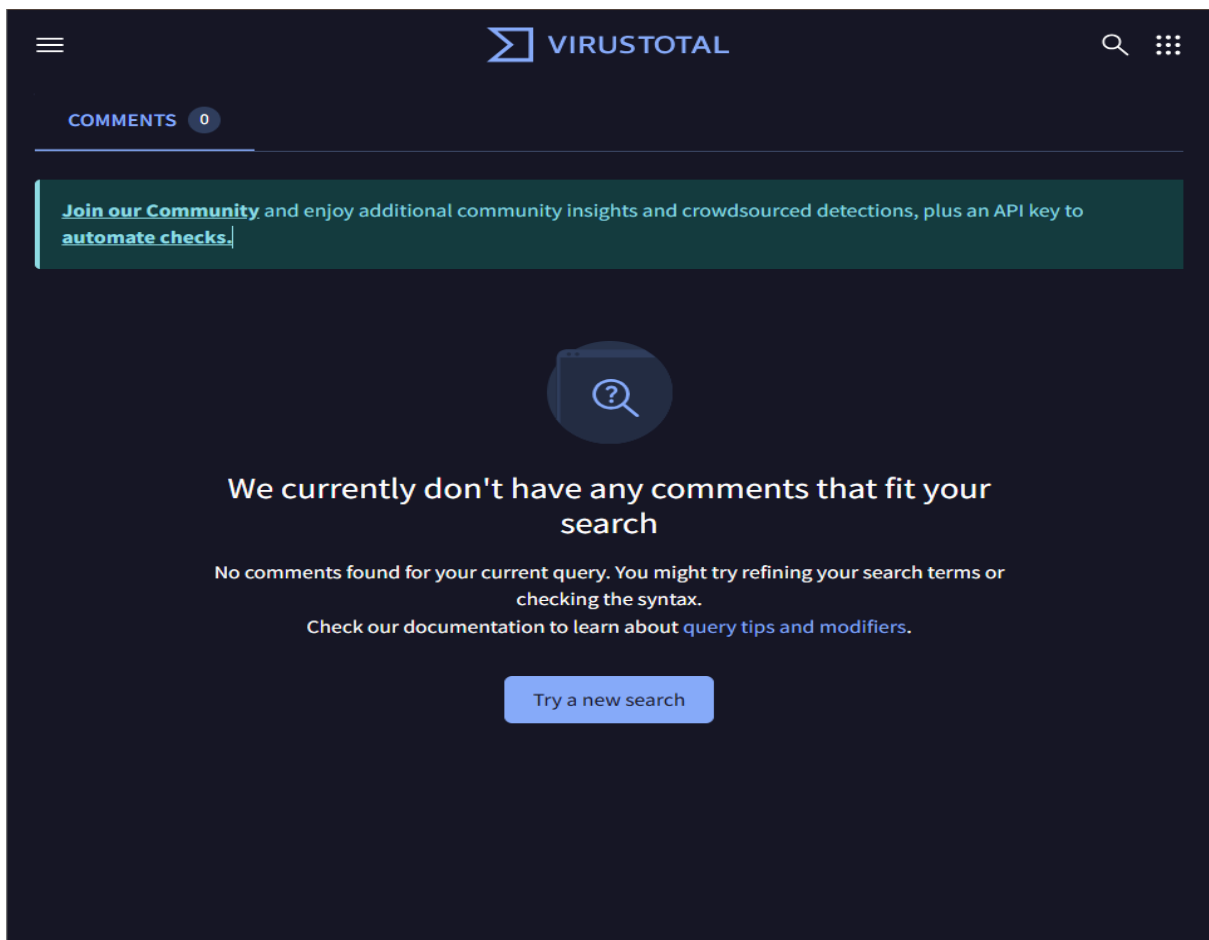


Figure 17 virustotal-hash

Indicator	Check	Result
IP 192.168.29.50	VirusTotal	Private RFC1918
File Hash	VirusTotal	Never submitted
Username/Hostname	AlienVault OTX	0 pulses
Kernel Version	6.17.0-14	

5.11 Root Cause Analysis

Immediate Cause:

Two SSH typos + legitimate finance activity within 300 seconds triggered correlation rule.

Why This is a False Positive:

Evidence	Finding
SSH Process ID	Same ID for all attempts - proves typos within single session

Evidence	Finding
7-day Baseline	Daily finance access, 3-4 weekly backups - routine behavior
2+ Year Backup Structure	Monthly backups since 2024 - established practice
File Integrity	SHA256 hashes identical - no data modification
Permission Denied Events	Security controls blocked sensitive files
Exfiltration Check	Zero file transfer commands
Persistence Check	No cron jobs, new SSH keys, or services
Lateral Movement Check	All connections from single source IP
Log Integrity	Logs intact - no tampering
Threat Intelligence	No hits - all artifacts legitimate

6. MITRE ATT&CK MAPPING

Tactic	Technique ID	Technique Name	Observed Activity
Initial Access	T1078	Valid Accounts	SSH authentication with valid credentials
Execution	T1059.004	Unix Shell	Command execution via bash
Privilege Escalation	T1548.003	Sudo and Sudo Caching	sudo -s to gain root privileges
Credential Access	T1110.001	Password Guessing	2 failed SSH attempts (typos)
Collection	T1005	Data from Local System	Access to financial documents
Collection	T1074.001	Local Data Staging	Files copied to /root/finance_backups/

7. IMPACT ASSESSMENT (NIST SP 800-61)

Impact Category	Rating	Reasoning
Confidentiality Impact	None	No unauthorized access; security controls blocked sensitive files
Integrity Impact	None	SHA256 hashes identical; no modifications
Availability Impact	None	Systems fully operational
Overall Business Impact	None	No disruption, no data loss

8. CONTAINMENT ACTIONS

Action Taken	Timestamp	Performed By	Verification
Ticket opened	11:13	Sai Prasad	Jira ticket SEC-OPS-2026-025 created
User contacted	12:21	Sai Prasad	User confirmed legitimate activity
No isolation required	N/A	Sai Prasad	No malicious indicators

9. ESCALATION DECISION

Field	Details
Escalated to L2 / IR Team	No
Reason for Escalation	False positive confirmed. No compromise indicators.

10. INCIDENT CLASSIFICATION

Field	Details
True Positive / False Positive	False Positive
Attack Success	N/A
Policy Violation	No

11. RECOMMENDATIONS

1. Implement a whitelist exception for authorized finance user sp from the internal network range 192.168.29.0/24 to reduce recurring false positives triggered by legitimate administrative activity.
2. Tune Wazuh correlation rule **100999** to include contextual filtering for known internal users performing authorized finance operations during standard business hours.
3. Establish a **change or activity tracking process for routine finance backup operations** so SOC analysts can quickly verify legitimate administrative tasks during investigations.
4. Enable **enhanced auditing and logging for sudo commands** to improve visibility into privileged activity and strengthen future incident investigations.
5. Deploy **File Integrity Monitoring (FIM) on critical finance directories** to automatically detect unauthorized modification of financial data and provide additional security monitoring.

12. TIMELINE OF KEY EVENTS

Time (IST)	Event
10:59:13	SSH authentication failure (attempt 1) – process ID 8064
10:59:20	SSH authentication failure (attempt 2) – same process ID
10:59:30	SSH authentication success (attempt 3) – same process ID
10:59:30 – 11:03:06	User begins finance session and navigates directories
11:03:06	Access to Tier-0 finance directory detected
11:03:10 – 11:03:18	User reviews financial files (permission checks and file read)
11:03:28	Sudo privilege escalation executed
11:03:32	Backup directory /root/finance_backups/jan_2026 created
11:03:34 – 11:03:39	Finance files copied to backup directory
11:03:39	Sensitive finance backup file activity detected
11:03:40 – 11:03:50	File permissions applied and session closed
11:03:51	Wazuh correlation rule 100999 triggered – CRITICAL multi-stage alert
11:13	SOC ticket SEC-OPS-2026-025 created and investigation begins
11:18	SSH authentication analysis confirms typo-based failures
11:21	Initial command history review completed
11:26	Investigation paused due to critical Windows security alert
11:27 – 12:26	Windows incident investigation and remediation performed
12:27	Ubuntu investigation resumed
12:30	Historical baseline analysis completed
12:33	Backup structure review completed
12:24:47	SHA256 verification confirms identical backup hash
12:35	Exfiltration and persistence checks completed – no indicators
12:37	Lateral movement and log integrity verification completed
12:38	Threat intelligence verification completed – no matches
12:39	Root cause analysis completed – determined false positive
12:40	Incident closed – FALSE POSITIVE

SLA Performance:

- Time to Triage: 2 minutes
- Active Investigation Time: 35 minutes
- Total Elapsed Time: 1 hour 25 minutes (with pause)
- SLA Target: 2 hours - MET

14. COMPLIANCE & FRAMEWORK ALIGNMENT

Framework	Phase / Coverage
NIST Incident Response Phases	Detection → Analysis → Containment → Eradication → Recovery
MITRE ATT&CK Coverage	T1078, T1059.004, T1548.003, T1110.001, T1005, T1074.001

15. LESSONS LEARNED

Aspect	Observation
What Worked Well	Process ID analysis proved typographical error definitively
What Worked Well	7-day historical baseline provided behavioral context
What Worked Well	SHA256 hash comparison eliminated data modification concerns
What Worked Well	2+ year backup structure proved established practice
Improvement Opportunity	Rule 100999 needs whitelist for known users
Improvement Opportunity	Change tickets needed for routine finance access
Improvement Opportunity	Kernel patch management needs improvement

16. CLOSURE SUMMARY

Field	Details
Final Verdict	False Positive - Legitimate Administrative Activity
User Compromised	No
Credentials Stolen	No
Data Exfiltrated	No
Persistence Installed	No
Lateral Movement	No
Files Modified	No
Affected Users	1 (targeted, not compromised)
Active Investigation Duration	35 minutes
Time from Alert to Closure	1 hour 25 minutes (with pause)
Business Risk Post-Incident	Low
Incident Closed By	Sai Prasad, SOC Analyst L1
Closure Date & Time	2026-03-09 12:40 IST